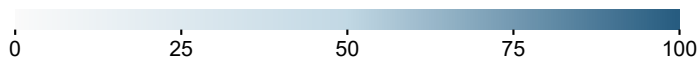


a Fire only**b Matched DeepSets****c Sum GNN**

	Terminal						Time average J						Terminal						Time average J						Terminal						Time average J														
	10%			25%			50%			10%			25%			50%			10%			25%			50%			10%			25%			50%			10%			25%			50%		
	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%									
IID (160)	96.9	98.1	99.8	96.7	96.8	99.7				98.8	99.6	99.9	97.7	99.4	99.8				99	99.5	99.9	98.1	99.4	99.8				99	99.5	99.9	98.1	99.4	99.8				99	99.5	99.9	98.1	99.4	99.8			
	98.8	99.9	99.3	96.8	99.1	99.5				99.3	99.9	99.5	98.4	99.6	99.8				99.5	99.9	99.6	98.4	99.7	99.9				99.5	99.9	99.6	98.4	99.7	99.9				99.5	99.9	99.6	98.4	99.7	99.9			
Count OOD (264)																																													
ISO 834 (158)	0	0	0	0	0	0				74	77.4	74.8	64.6	82.5	84.3				73.2	75.4	73.9	69.4	79	83.5				73.2	75.4	73.9	69.4	79	83.5				73.2	75.4	73.9	69.4	79	83.5			
ASTM E119 (95)	0	0	0	0	0	0				75.7	78.4	80.6	74.1	82.7	88				81.1	81.5	87.1	76.7	82.2	91.6				81.1	81.5	87.1	76.7	82.2	91.6				81.1	81.5	87.1	76.7	82.2	91.6			
Perturbed ISO (111)	34.4	26.4	16.7	6.4	17	17.7				74.2	68.8	82.8	75.3	79.9	81.7				77.4	74.2	81.9	82.1	81.6	87.4				77.4	74.2	81.9	82.1	81.6	87.4				77.4	74.2	81.9	82.1	81.6	87.4			
Plateau (103)	98.7	98	99.8	98	92.1	98.9				98.7	99.3	99.9	97.3	98.9	100				99.4	99.6	100	98.2	98.8	99.8				99.4	99.6	100	98.2	98.8	99.8				99.4	99.6	100	98.2	98.8	99.8			
Decay (113)	90.2	96.2	99.7	79.3	87.2	98.5				97.1	98.1	99.7	96.5	95.7	99				97.4	98.7	99.6	95.9	96.6	99.6				97.4	98.7	99.6	95.9	96.6	99.6				97.4	98.7	99.6	95.9	96.6	99.6			
Log variant (69)	100	98.8	99.6	99.8	98.3	99.3				97.1	100	99.7	99.9	100	100				97.2	100	99.8	99.7	100	100				97.2	100	99.8	99.7	100	100				97.2	100	99.8	99.7	100	100			
External fire (49)	0	0	0	0	0	0				72	81.2	83.6	73.5	79.9	93				79.2	84	89.5	87.6	83.9	88.8				79.2	84	89.5	87.6	83.9	88.8				79.2	84	89.5	87.6	83.9	88.8			
Bilinear (77)	98.6	94.3	90.2	96.1	96.4	92.1				98	98.1	92.5	99.1	96.5	92.3				96.5	96	89.5	99.1	97.9	93.5				96.5	96	89.5	99.1	97.9	93.5				96.5	96	89.5	99.1	97.9	93.5			
Linear (125)	99.5	97.5	98.6	99.3	98.5	99				100	99.4	99.4	100	99.2	99.6				100	99.2	99.5	100	99.4	99.7				100	99.2	99.5	100	99.4	99.7				100	99.2	99.5	100	99.4	99.7			
Smoldering (97)	86	100	100	99	100	100				81.1	100	100	94.5	100	100				79.3	100	100	94.5	100	100				79.3	100	100	94.5	100	100				79.3	100	100	94.5	100	100			

Five-seed mean; all 12 protocols × 2 endpoints × 3 case-slot budgets



Available oracle improvement captured (%)

0 = uniform-selection expectation; 100 = retrospective oracle. Gray / NA = undefined ratio.

Expected cutoff-tie selection. Post-review exploratory analysis; no risk or time-saving claim.